Update: A Claimed $\mathcal{O}(2^{0.49n})$ Algorithm for Subset Sum

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Abstract

The accompanying survey, *A Survey of Exact Algorithms for Subset Sum and Its Variants*, described a fixed-exponent worst-case improvement over $2^{n/2}$ as open. On October 6, 2026, OpenAI released a manuscript collection containing a preprint dated October 4 that claims a uniform randomized classical $\mathcal{O}(2^{0.49n})$ decision algorithm for positive-integer Subset Sum. This note records the exact scope of that claim, its five-stage proof pipeline, and what has been independently machine-checked in Lean versus what remains unverified. It is an audit update, not an independent proof of the claimed runtime bound.

1 Why this update is separate

The earlier survey remains useful as a record of the literature it described, but several of its central statements are now historically dated. In particular, its introduction, result table, chronology, conclusion, and open-problem discussion state that ordinary worst-case Subset Sum had no known constant improvement below $2^{n/2}$. Rewriting each statement in place would blur the distinction between established results and a new, not yet independently verified claim. This dated note supplies the qualification.

2 The claimed result

The primary source is the preprint *Subset Sum in Time* $\mathcal{O}(2^{0.49n})$ [1], included in OpenAI's mathematics collection [2]. Its Theorem 1.1 claims a single uniform randomized classical word-RAM algorithm with the following guarantees.

- The input is a target $t \geq 0$ and positive integer weights $a_1, \ldots, a_n$, with repeated weights and the empty subset allowed.
- Let b be the largest input bit length. The word length is $w = \lceil 4(n + b + \log_2(n + 2)) \rceil$ bits. Random word generation, input access, preprocessing, and writable storage all count toward the running time.
- The algorithm halts on every random execution and returns the correct yes/no decision answer with probability at least $2/3$ for every fixed input.
- For every fixed $c > 0$, inputs with $b \leq n^c$ are processed in $C_c 2^{0.49n}$ word operations for sufficiently large n, on every random execution. The stated intermediate form is $2^{0.489995n}\,\mathrm{poly}(n + b + 2)$.

Table 1: Status of the main classical regimes.

Setting	Bound	Status
Positive-integer Subset Sum, polynomial-bit word RAM	$2^{0.49n}$	claimed in a 2026 preprint
Signed-integer Subset Sum with occurrence witness	$2^{n/2}$ baseline	reductions machine-checked in Lean; $2^{0.49n}$ core unformalized
Random hard knapsacks	$2^{0.291n}$	BCJ model; average-case
Pigeonhole Equal Subset Sum	$2^{n/3}$	randomized promise result

If verified, the theorem changes the worst-case decision landscape. Note that Theorem 1.1 states a positive-integer decision result rather than a signed-integer witness-recovery theorem; reducing signed inputs to positive weights and lifting a decision oracle to an explicit subset witness require separate reductions.

3 Pipeline in the manuscript

The new proof is not a direct extension of the Howgrave-Graham–Joux or Becker–Coron–Joux representation algorithm. Its five coupled stages are:

1. **Isolation.** Random tie-breaking weights $u_1, \ldots, u_n$ transform one input (a, t) into polynomially many positive-weight targets $a'_i = \text{scale} \cdot a_i + u_i$ and $t'_s = \text{scale} \cdot t + s$. With constant probability, a satisfiable input produces a target t'_s with a unique representing subset.

2. **Preliminary search.** A capped meet-in-the-middle procedure solves instances whose random restrictions have a sufficiently small set of distinct subset sums.

3. **Modular checksum.** For a large random prime p, a filtered high/low-digit decomposition ($a'_i = q_i B + r_i$) estimates a Fourier expression for each modular target bin to small *absolute* error. The filter uses fixed numerical parameters and a finite interval-arithmetic certificate in the appendix.

4. **Alias extraction.** A three-block shared-mask representation lists modular false positives (subsets with $a' \cdot x \neq t'_s$ but $a' \cdot x \equiv t'_s \pmod p$). Records carry masks on the shared block and are compatible precisely when those masks are disjoint.

5. **Subtraction and thresholding.** Subtracting the enumerated alias contribution from the estimated modular bin count separates an empty bin from an isolated solution.

The claimed exponent depends on the interaction of checksum variance, filtering coverage, alias enumeration, finite precision, and explicit caps on every randomized loop.

4 Verification status

The official Lean formalization catalog accompanying the collection does not list this manuscript [3], and the release notes state that the manuscripts are at varying stages of verification [2]. We therefore

treat the $2^{0.49n}$ runtime bound as *unreviewed until independently checked*.

In our accompanying Lean 4 verification layer, the deterministic structural skeleton of the five-stage pipeline has been formalized and machine-checked:

- the signed-to-nonnegative and strictly positive weight reductions with an involutive witness map;
- the tie-breaking isolation family $a'_i = \text{scale} \cdot a_i + u_i$ and $t'_s = \text{scale} \cdot t + s$, including decision equivalence and witness uniqueness when a tie sum s is isolated;
- the capped distinct-sum preliminary search and the linear high/low-digit subset-sum split;
- the three-block shared-mask compatibility solver, the exact decomposition of each modular target bin into true solutions and modular aliases, and the end-to-end decision equivalence after alias subtraction.

What remains unformalized is the quantitative probabilistic core: the filtered Fourier checksum variance bound, the random-prime alias-survival cap, the rational interval-arithmetic certificate in Appendix A, and the word-RAM operation count.

5 Next verification steps

1. Formalize and check the filtering-moment inequalities and Appendix A's rational interval certificate.
2. Verify the randomized Fourier checksum estimator and capped alias extractor bounds.
3. Update the main survey's worst-case table from "open" only after the probabilistic and word-RAM bounds are independently established.

References

[1] OpenAI, "Subset Sum in Time $\mathcal{O}(2^{0.49n})$," October 4, 2026. `https://github.com/openai/math/blob/main/preprints/Subset-Sum-in-Time-2-power-0-49n-October-4-2026/subset-sum.pdf`

[2] OpenAI, "Sharing AI progress in mathematics," October 6, 2026. `https://openai.com/index/sharing-ai-progress-in-mathematics/`

[3] OpenAI, "Lean formalization catalog," accessed October 9, 2026. `https://github.com/openai/math/blob/main/lean/formalization.yaml`

[4] A. Becker, J.-S. Coron, and A. Joux, "Improved generic algorithms for hard knapsacks," EUROCRYPT 2011. `https://eprint.iacr.org/2011/474`

[5] C. Jin, R. Williams, and S. Zhang, "New algorithms for Pigeonhole Equal Subset Sum," ESA 2025. `https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ESA.2025.86`